

Explaining the Paper Simply

Why Japan's incineration energy-recovery pattern matters

8-10 MINUTE LIVE ROUTE

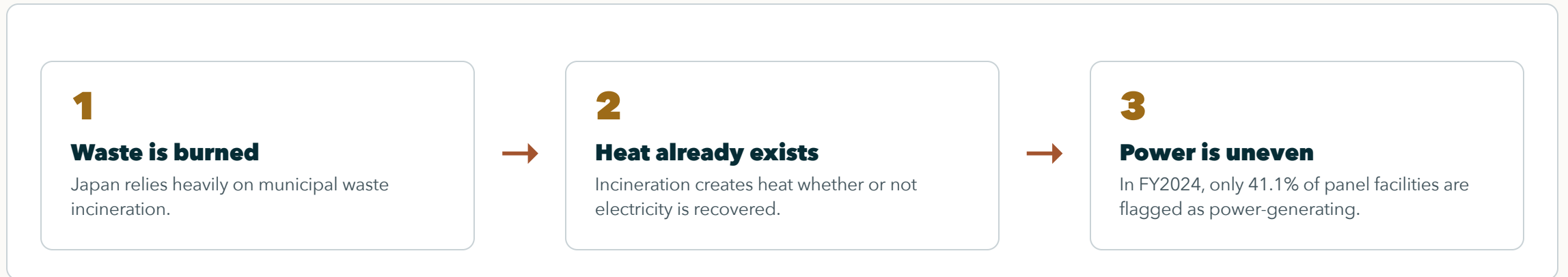
PAPER ONLY

APPENDIX ONLY IF ASKED

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Paper briefing for discussion and feedback

Why This Matters



The practical question is simple: where does useful electricity recovery appear, and where does it not?

The Paper's Main Claim

ONE SENTENCE

Japan's incineration fleet is a two-part modernization problem, not one average transition curve.

PROBLEM 1

Some plants do not enter electricity generation.

This is the entry problem.

PROBLEM 2

Generators do not perform equally well.

This is the performance problem.

What the Paper Compares

NON-GENERATORS

Who first reports power generation?

This identifies where entry into energy recovery appears.

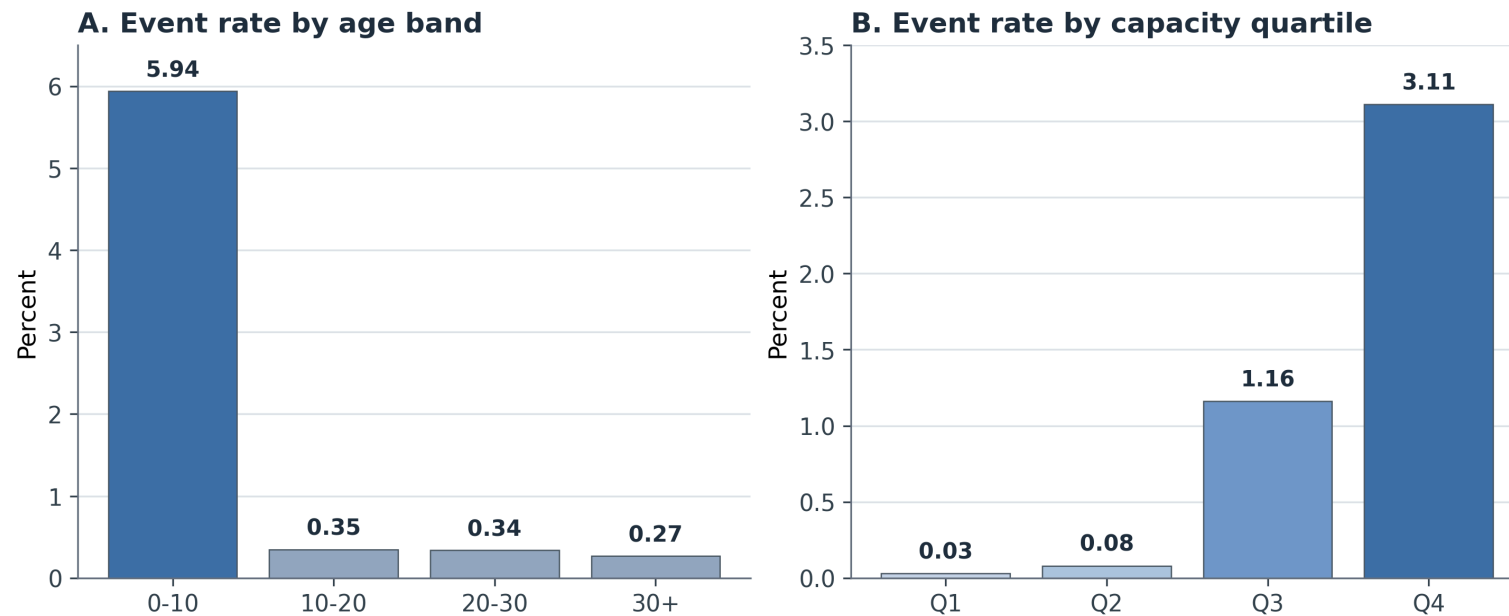
GENERATORS

Who recovers more electricity per tonne?

This identifies how uneven performance is after entry.

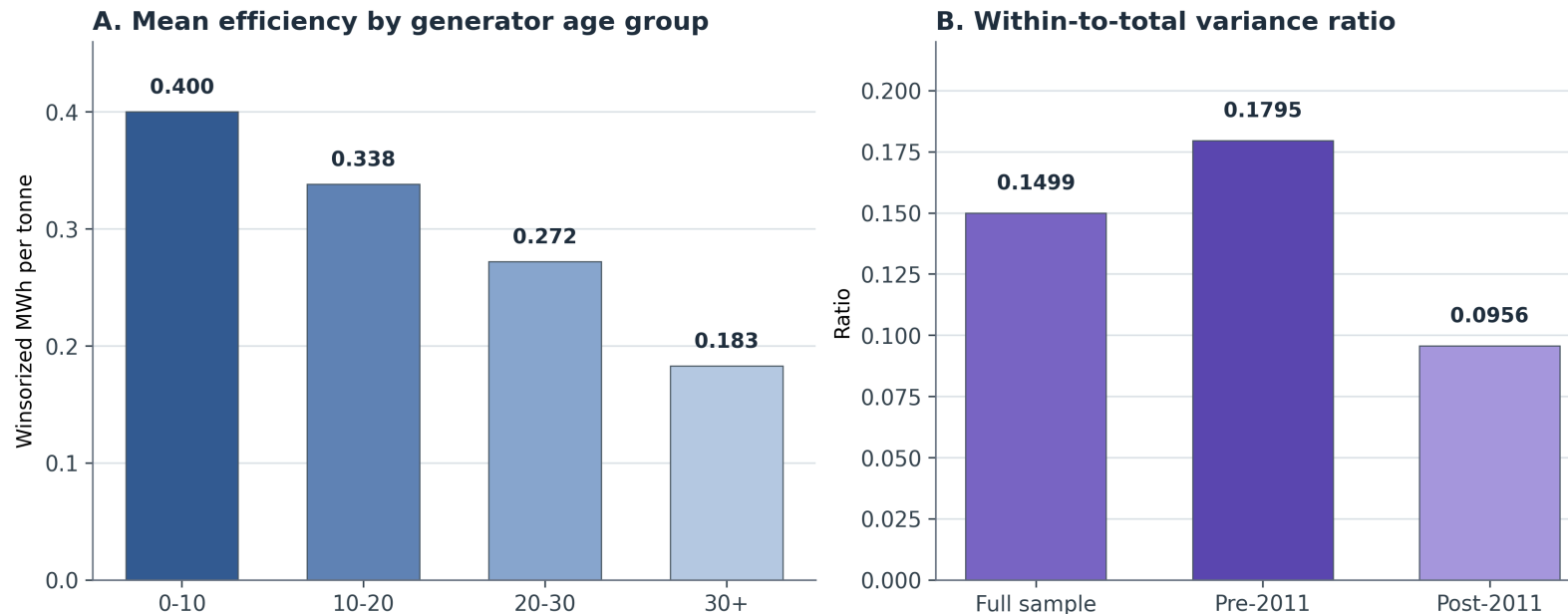
The evidence is national facility-level data from FY2005-FY2024, not a small case study.

Result 1: Entry Is Selective



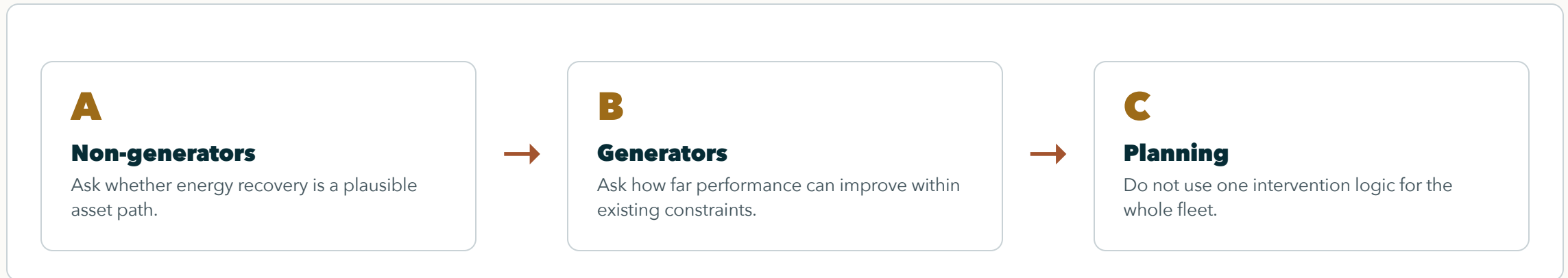
Entry into power generation is concentrated among younger and larger facilities: 102/141 events are from age 0-10 facilities, and 99/141 are in the largest capacity quartile.

Result 2: Performance Is Uneven



Among generators, older facilities recover less electricity per tonne. Scale and utilization help, but they do not erase large facility-level differences.

What Changes If We See Two Problems?



This matters because renewal, entry, and generator optimization are different decisions.

Why This Can Stand as a Paper

CLEAR CONTRIBUTION

It separates entry from performance.

That avoids a misleading one-average story.

USEFUL EVIDENCE

It uses one national facility panel.

The same fleet is read across two linked margins.

DEFENSIBLE CLAIM

It stays diagnostic.

The paper shows where bottlenecks appear without overstating causality.

What the Paper Does Not Claim

BOUNDARIES

- No strict causal policy-effect claim.
- No proof that replacement is the only pathway.
- No complete lifecycle climate accounting.
- No claim that the generator frame is a perfect census.

DEFENDED CLAIM

This is a diagnostic fleet decomposition.

It shows two bottlenecks: selective entry into generation and uneven performance among generators.

Feedback Ask

The paper's message is: entry first, performance second.

Useful feedback: is this two-part story clear, defensible, and worth using as the paper's main pitch?

Appendix A: Model Details

ADOPTION MODEL

Lagged discrete-time logit hazard among facilities still at risk of first observed generation.

- Outcome: first report of power generation in the next observed year.
- Predictors: prior-year age band and design capacity.
- Controls: fiscal-year fixed effects and prefecture fixed effects.
- Uncertainty: facility-clustered standard errors.

EFFICIENCY MODEL

Descriptive panel models among identifiable operating generators.

- Outcome: winsorized log electricity generated per tonne processed.
- Predictors: age, capacity, utilization, heating value, grid-emission control.
- Models: pooled OLS, year FE, RE, and year FE plus RE.
- Uncertainty: facility-clustered standard errors.

Appendix B: Key Numbers

Evidence item	Number	Interpretation
FY2024 facilities flagged as power-generating	41.1%	Most facilities remain outside electricity generation in the panel.
Source panel	23,599	Facility-year observations across FY2005-FY2024.
Identifiable facilities	2,948	National facility-level evidence.
Adoption risk-set size	13,770	Facility-years first observed without generation.
Observed first-adoption events	141	Events used to describe entry into generation.
Events from age 0-10 facilities	102	Adoption is concentrated among young facilities.
Events in largest capacity quartile	99	Adoption is concentrated among large facilities.
Generator regression frame	5,683	Identifiable operating-generator observations.
Within-to-total variance ratio	0.1499	Most generator-efficiency variation is between facilities.

Appendix C: Pathway Audit

RESET / REBUILD-LIKE

82

observed events

CONTINUITY / UPGRADE-LIKE

38

observed events

PLACEHOLDER / FORWARD-DATED

20

observed events

Safe wording: selective modernization with capital-intensive evidence, not proof that replacement is the only pathway.

Appendix D: File Map

SLIDES

`paper/slides/paper-zoom-briefing.md`

Editable source deck.

SCRIPT

`paper/slides/paper-zoom-script.md`

Separate live presentation script.

CHECKLIST

`paper/slides/paper-zoom-presentation-checklist.md`

Run sheet for a simple, direct presentation.

BUILD COMMAND

`npm run slides:paper:pdf`

Regenerates the shareable PDF.